

Junwei Quan

Toronto, Ontario | jquan287619461@gmail.com | github.com/Wei-cyber

Professional Summary

Computer Science and Psychology graduate with experience in machine learning research, geospatial data analysis, and digital communications. Skilled in building data-processing pipelines, conducting technical research, and collaborating across interdisciplinary teams. Strong analytical, communication, and problem-solving abilities developed through research and leadership experience.

Education

York University — Toronto, Ontario

Bachelor of Science (B.Sc., Hons), Computer Science | May 2026

Bachelor of Science (B.Sc.), Psychology | Aug 2023

Experience

York University — Research Assistant

May 2024 – Apr 2025

- Researched and evaluated machine learning techniques for remote sensing image segmentation in Python.
- Built end-to-end pipelines for processing field measurements and LiDAR datasets.
- Generated annotated datasets and supported model development for understory vegetation mapping.
- Conducted literature reviews and collaborated with researchers on data analysis methodologies.

B-minor Compiler

C, Flex, Bison, Assembly, GNU Make

- Reduced onboarding time for peers by creating documentation describing key workflows and module ownership within a codebase.
- Participated in code reviews, incorporating feedback to improve code quality, readability, and adherence to team standards.
- Leveraged Flex with regular expressions for scanner generation and Bison for parser generation.
- Constructed and traversed ASTs to support type checking and code generation.

Technical skills

Programming: Python, Java, SQL, C, JavaScript

Data & ML: Machine Learning Pipelines, Data Analysis, Data Annotation, Remote Sensing

Tools: Git, Microsoft Office, Google Workspace

Additional: Communication, Team Collaboration, Problem Solving, Time Management